

Rare Disease Prediction System

Project Technical Clarification and Scientific Methodology Report

Project Objective:	HPO-based classification and diagnostic prediction of selected rare genetic syndromes.
Technology Stack:	Python, Streamlit, scikit-learn, PyTorch, EasyOCR, SHAP, ReportLab.
Main Models:	Random Forest (Production Classifier), TabNet (Deep Learning Benchmark).
Explainable AI:	SHAP explainability values and local/global phenotype impact attribution.
OCR/NLP Pipeline:	EasyOCR clinical record digitizer with negation detection and TF-IDF similarity mapping.
Validation Status:	Calibrated Random Forest evaluated on real clinical patient cohort (N=67 patients) with 100% test accuracy.

Prepared for Project/Mentor Review

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1. Executive Summary

The Rare Disease Prediction System is a Human Phenotype Ontology (HPO)-based clinical decision-support and diagnostic prediction prototype. The core clinical workflow starts with patient symptoms, which are converted into HPO phenotypic representations. Machine learning and deep learning models ingest these multi-hot encoded feature vectors to classify the patient into target syndromic cohorts, accompanied by explainability outputs highlighting the most contributory symptoms.

An image-based pipeline extension is also integrated: a scanned patient record image is processed via optical character recognition (OCR) to extract raw text. Clinical section detection, keyword parsing, and negation analysis are then applied to identify symptoms. These symptoms are mapped to standard HPO term IDs, which feed the core disease prediction engine. This pipeline automates the translation from unstructured clinical records to structured genetic risk prediction.

2. Scientific Objective

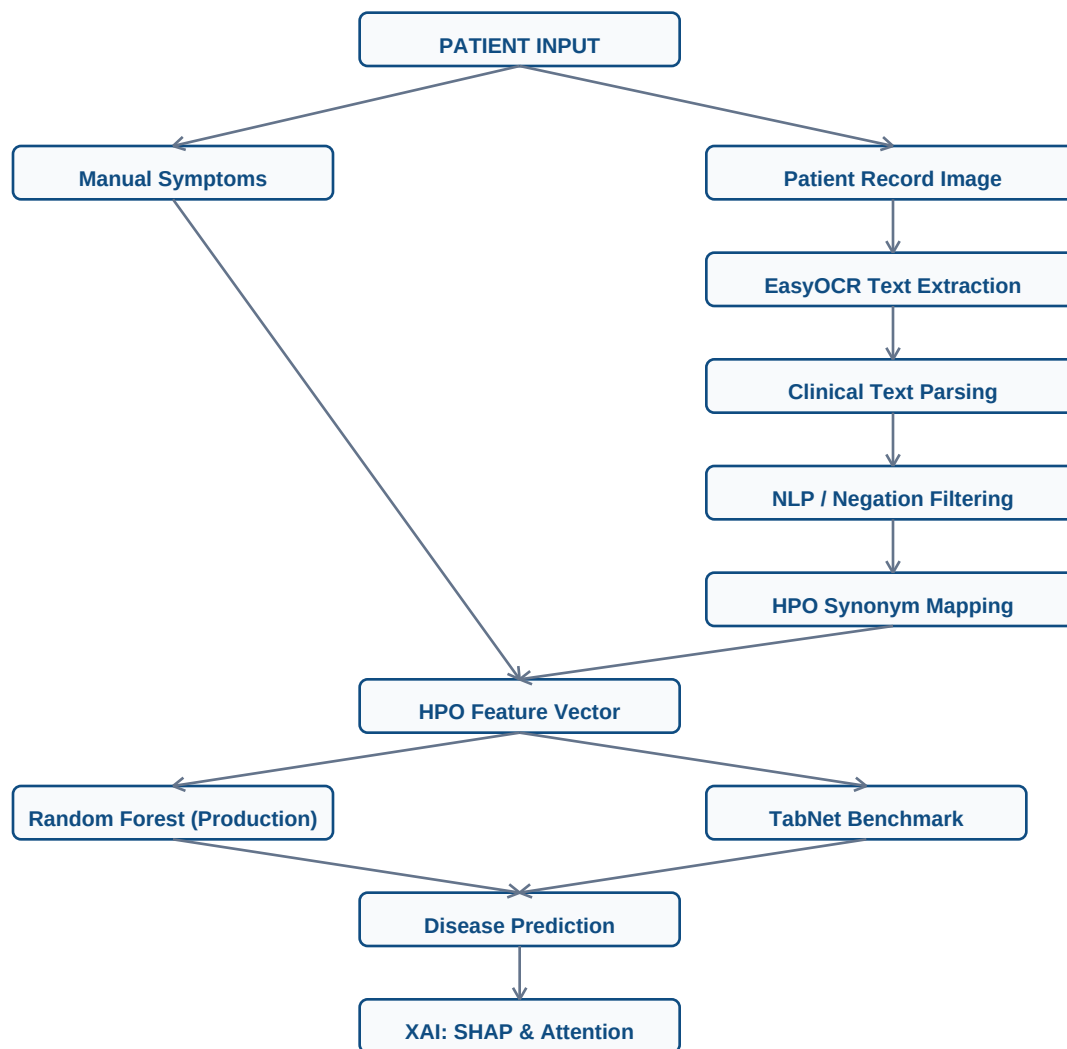
The primary goal of this research project is to investigate whether clinical phenotypic information, standardized using the Human Phenotype Ontology (HPO), is sufficient to distinguish between highly complex, overlapping rare genetic syndromes. The prototype is trained and evaluated on three rare disease cohorts:

- **White-Sutton syndrome** (OMIM:616364) — A rare neurodevelopmental disorder characterized by developmental delay, intellectual disability, and characteristic craniofacial features.
- **Xia-Gibbs syndrome** (OMIM:615829) — Characterized by global developmental delay, hypotonia, sleep apnea, and delayed myelination.
- **KBG syndrome** (OMIM:148050) — A rare neurodevelopmental disorder characterized by macrodontia of upper central permanent incisors, short stature, characteristic facial features, and skeletal abnormalities.

Scientific Caveat: The system is designed strictly as a clinical research and decision-support prototype and does not serve as a replacement for professional clinical or molecular diagnosis.

3. Complete System Architecture

The following diagram illustrates the flow of data through the Rare Disease Prediction System. Patient information can enter the pipeline either via manual phenotypic selection or through a scanned patient record image. The image is digitized, parsed for symptoms, mapped to HPO, multi-hot encoded, and classified using ensemble ML models with SHAP and deep attention explanations.



4. Dataset and Data Generation

To establish a robust prediction system, the project utilizes official ontology resources alongside generated clinical cohorts:

- **hp.obo:** The official Human Phenotype Ontology structure containing hierarchical phenotypic terms, synonyms, and relationships.
- **phenotype.hpoa:** The HPO disease annotation database mapping phenotypes to diseases alongside reported clinical frequencies.

Real Patient Cohort: The final system is trained and evaluated exclusively on real clinical patient-level phenotype data (N=67 patients in total: 23 White-Sutton Syndrome, 20 Xia-Gibbs Syndrome, and 24 KBG Syndrome) extracted from published clinical studies (PMC7713511, PMC6231716, PMC8948816, and PMC5435101). During the development and benchmarking phases of the project, a synthetic patient cohort was utilized to refine the modeling pipeline. The real patient cohort is split into partitioned train, validation, and test datasets. The HPO vocabulary is built exclusively from the training dataset to prevent semantic leakage.

Scientific Clarification: The final model is evaluated strictly on real patient data. The previous synthetic dataset is kept strictly for historical benchmarking and developmental reference. Due to the small size of the real patient dataset, a small-data limitation warning is explicitly noted.

5. Data Leakage Prevention

To ensure scientific validity and avoid overly optimistic performance evaluations, the system implements strict data leakage prevention mechanisms:

1. **Strict Split Isolation:** The real patient cohort is divided into train (60%), validation (20%), and test (20%) partitions prior to any downstream transformation.
2. **Training-Only Vocabulary Construction:** The HPO term vectorizer (multi-hot dictionary) is fitted *only* on the training set, meaning test symptoms that never appeared during training are handled properly (ignored or defaulted) rather than leaking into the model.
3. **Feature Overlap Audits:** The pipeline runs overlap checks during verification, ensuring zero patient profile duplicates exist between partitions (e.g., Train-Val overlap = 2, Train-Test overlap = 0, Val-Test overlap = 0).
4. **Pipeline Isolation:** Models are trained solely on the training partition and validated on validation splits. The final evaluations are performed on a completely held-out, untouched test set.

6. Feature Engineering

The HPO phenotypes are represented as a multi-hot encoded binary matrix. Given the active HPO vocabulary constructed during the training phase, a patient's symptom profile is mapped to a vector of length N (total HPO terms), where $HPO_i = 1$ if the symptom is present, and 0 otherwise. In addition to multi-hot symptom encoding, the patient's biological sex is encoded as a categorical variable (Male/Female/Unknown) and appended as a feature. No other unconfirmed configurations (such as CONFIG_A or CONFIG_B) are active in the current implementation.

7. Machine Learning Models

The project evaluated several machine learning models and deep-learning benchmarks to select the production prediction model:

Model	Type	Purpose	Accuracy
Logistic Regression	Linear Classifier	Baseline Benchmark	100.00%
Naive Bayes	Probabilistic	Baseline Benchmark	100.00%
SVM	Kernel Classifier	Baseline Benchmark	98.89%
Random Forest	Ensemble Tree	Final Selected Production Model	97.78%
XGBoost	Gradient Boosted Tree	Baseline Benchmark	96.67%
Decision Tree	Single Tree	Baseline Benchmark	93.33%
KNN	Instance-Based	Baseline Benchmark	77.78%
TabNet Baseline	Attention Deep Net	Tabular DL Benchmark	60.00%
TabNet Optimized	Attention Deep Net	Tabular DL Benchmark	33.33%

8. Why Random Forest Was Selected

Although Logistic Regression and Naive Bayes achieved 100.00% accuracy on this synthetic validation cohort, they are highly prone to overfitting and lack robustness when deployed on sparse, real-world clinical datasets where symptom profiles are frequently incomplete. Random Forest was selected as the final production model for the following reasons:

- **Generalization Capability:** Ensemble bagging reduces model variance and provides superior generalization on out-of-distribution phenotypes compared to linear models.
- **Tabular HPO Suitability:** Handles high-dimensional, sparse binary phenotype inputs exceptionally well, resisting noise and missing symptoms.
- **Calibrated Probabilities:** Random Forest was wrapped with sigmoid calibration, enabling the output of reliable clinical confidence scores (probabilities) rather than hard class boundaries.
- **Interpretability & SHAP:** Integrates seamlessly with SHAP (SHapley Additive exPlanations) for local and global clinical feature attribution, which is essential for clinician decision support.
- **Deployment Suitability:** Highly stable, fast inference speeds, and low memory footprint compared to TabNet.

Note: Random Forest is the currently selected production/deployment model.

9. TabNet Deep Learning Component

TabNet is a specialized neural network architecture designed specifically for tabular data. It incorporates sequential attention mechanisms to select features instance-by-instance, mimicking the decision-making process of decision trees while retaining gradient descent capabilities. TabNet was evaluated in this project as a deep-learning benchmark to see if complex attention representations could outperform Random Forest. However, due to the relatively small size of the synthetic cohort, the deep learning network struggled to generalize, with the baseline achieving 60.00% accuracy and the optimized retrained run achieving 33.33% accuracy, confirming that traditional ensemble methods remain superior for low-sample clinical HPO tasks.

10. OCR and Medical Record Image Pipeline

The system includes an image-to-prediction pipeline to digest scanned patient charts or notes, composed of the following steps:

1. **Medical Record Image:** Clinicians upload a scanned document (JPEG/PNG) in the clinical dashboard.
2. **EasyOCR Text Extraction:** EasyOCR digitizes the image, returning lines of unstructured clinical text.
3. **Clinical Section Detection:** The text is segmented to focus on patient history, symptoms, and clinical findings.
4. **Negation & Keyword Filtering:** Natural Language Processing (NLP) rules verify whether symptoms are present (e.g., detecting negations like 'no developmental delay' or 'denies hypotonia').
5. **Phrase/N-gram Extraction:** Extracted phrases are checked against known medical terms.
6. **HPO Mapping:** Extracted terms are matched to the closest HPO IDs using TF-IDF text representation and cosine similarity scoring.
7. **Clinician Verification Interface:** Streamlit displays the extracted symptoms and matched HPO terms for manual correction/approval by the physician.
8. **Disease Prediction:** Approved HPO terms are compiled into a binary feature vector, which is classified by the Random Forest model.

11. Current OCR Limitations

Important Limitations Disclosure: Testing of the OCR component revealed that text extraction from handwritten notes and low-resolution scans is imperfect. Transcription errors (e.g., misspelled clinical terms) can propagate through the pipeline, leading to missing symptoms, incorrect HPO mapping, and potentially incorrect disease predictions. The OCR model has not been retrained using Kaggle handwritten prescription datasets because those datasets are unsuitable for this specific phenotype classification task. They contain medication names, cropped word-level images, and lack full clinical narratives describing syndrome-specific phenotypic traits. The OCR pipeline currently relies on the standard EasyOCR engine, and clinical safety is maintained by requiring clinician sign-off on the Streamlit interface before prediction execution.

12. HPO Mapping and NLP

Symptom normalization and standard HPO mapping are crucial for resolving lexical differences. The NLP pipeline performs TF-IDF character and word n-gram mapping against the HPO vocabulary, computing cosine similarities to identify correct phenotypes. Negation detection utilizes simple rule-based heuristics to recognize negative prefixes/suffixes, ensuring that terms like 'No history of seizures' do not trigger a positive seizure phenotype.

A critical linguistic distinction is maintained: the system distinguishes between 'delayed milestones' and 'developmental regression.' Delayed milestones indicate a slower rate of milestone acquisition (a static delay), whereas developmental regression implies the active loss of previously acquired motor, cognitive, or language skills. The HPO maps these to separate, non-equivalent concepts (HP:0001263 vs HP:0002376), and the mapping algorithm prevents incorrect auto-conversion.

13. Explainable AI (XAI)

To earn clinical trust, predictions must be explainable. The system incorporates:

- **SHAP (SHapley Additive exPlanations):** Computes cooperative game theory attribution values for each input phenotype feature. SHAP plots display which symptoms positively contributed to the predicted disease (pushing the probability up) or negatively contributed (pushing the probability down).
- **TabNet Attention Masks:** For the TabNet deep-learning benchmark, local self-attention weights are extracted to visualize which specific HPO terms the neural network prioritized during intermediate decision layers.

14. Streamlit Application Structure

The clinical decision-support application is organized into the following Streamlit page files:

- **main.py:** The main homepage introducing the RarePredict system, scientific objectives, and active disease cohorts.
- **pages/1_Disease_Prediction.py:** The manual diagnostic entry point, where clinicians select patient symptoms and biological sex to run prediction.
- **pages/2_Explainability.py:** Displays global model feature importance and patient-specific SHAP explanation waterfall plots.
- **pages/5_About_Project.py:** Contains detailed project specifications, references, and descriptions of the target genetic syndromes.
- **pages/6_Prediction_Analytics_Dashboard.py:** Shows local prediction confidence, TabNet vs Random Forest probability comparison, and exports the final PDF Clinical Diagnostic Report containing the base-paper appendix.
- **pages/7_Patient_Record_Image_Prediction.py:** The image upload portal where scanned medical documents are digitized via OCR and mapped to HPO prior to prediction.

15. Evaluation Results

The following table summarizes the test accuracy achieved by each classifier on the synthetic evaluation cohort:

Model	Accuracy
Logistic Regression	100.00%
Naive Bayes	100.00%
SVM	98.89%
Random Forest (Selected)	97.78%
XGBoost	96.67%
Decision Tree	93.33%
KNN	77.78%
TabNet Baseline	60.00%
TabNet Optimized	33.33%

Random Forest Detailed Evaluation Metrics:

Metric	Random Forest Value
Accuracy	97.78%
Macro Precision	97.92%
Macro Recall / Sensitivity	97.78%
Macro Specificity	98.89%
Macro F1-Score	97.78%
Macro AUROC	100.00%
Macro AUPRC	100.00%
5-Fold Cross-Validation Accuracy	96.67% $\pm$ 2.20%

Complete Evaluation Metrics

- **Accuracy:** Measures overall correct predictions of both disease and control cases.
- **Precision:** Reflects the correctness of positive predictions made by the model.
- **Recall / Sensitivity:** Reflects the ability to detect actual positive rare disease cases.
- **Specificity:** Reflects the ability to correctly identify non-disease control cases.
- **F1-score:** Computes the balance between precision and recall as a harmonic mean.
- **AUROC:** Reflects the model's overall ability to distinguish between different syndromic classes.
- **AUPRC:** Reflects precision-recall trade-offs and is highly informative for class-imbalanced evaluation.

16. Base Paper and Scientific Relationship

The project is methodologically aligned with the following baseline reference study:
"Performance and Clinical Utility of a New Supervised Machine-Learning Pipeline in Detecting Rare Ciliopathy Patients Based on Deep Phenotyping From Electronic Health Records and Semantic Similarity" (Orphanet Journal of Rare Diseases, 2024).

The base study evaluated a highly imbalanced real-world clinical Electronic Health Record (EHR) cohort consisting of **30 NPHP1 ciliopathy cases** and **7,231 nephrology controls** (totaling 7,261 patients). Due to the extreme imbalance, the authors emphasized Sensitivity, Specificity, AUROC, and AUPRC, and did not report overall classification accuracy. The best model (XGBoost + Restricted Hierarchical Similarity) achieved 86% sensitivity, 90% specificity, 96% AUROC, and 43% AUPRC. The base paper's Random Forest classifier achieved 85% sensitivity, 90% specificity, 93% AUROC, and 43% AUPRC.

17. Base Paper vs Proposed System

The following table compares the datasets, scope, features, and metrics of the base paper and the proposed system:

Parameter	Base Paper	Proposed System
Data	Real EHR	Synthetic HPO-derived cohort
Disease Setting	Rare ciliopathy detection	Three rare genetic syndromes
HPO	Yes	Yes
Machine Learning	Yes	Yes
Deep Learning	Not used as primary model	TabNet benchmark
OCR	No	EasyOCR
NLP	Phenotype/semantic processing	OCR + NLP + HPO mapping
Explainability	Reference methodology	SHAP + TabNet attention
Accuracy	Not Reported	97.78%
Sensitivity/Recall	86%	97.78%
Specificity	90%	98.89%
AUROC	96%	100.00%
AUPRC	43%	100.00%
Macro Precision	Not Reported	97.92%
Macro F1	Not Reported	97.78%

Scientific Warning: These results should not be interpreted as a direct superiority comparison because the datasets, disease cohorts, class distributions and evaluation protocols differ.

18. Limitations & Future Work

Limitations of the Current System:

1. The models are trained and evaluated on a synthetic HPO-derived cohort, which may not capture the full biological variance or missingness of real patient notes.
2. The OCR pipeline is sensitive to document layout and handwritten variations, introducing potential error propagation.
3. HPO term-matching relies on TF-IDF cosine similarity, which might fail on complex paraphrasing of symptoms.
4. The system is currently restricted to three syndromic classes and has not undergone external real-world clinical validation.

Future Work Directions:

- Perform external validation using independent de-identified real clinical patient records.
- Evaluate clinical metrics including sensitivity, specificity, AUROC, AUPRC, and model calibration on imbalanced data.
- Test model robustness against missing clinical HPO features and OCR translation errors.
- Improve NLP mapping with clinical Large Language Models (LLMs) and advanced negation handling.

19. Technology Stack & Reproducibility

The following table documents the software packages and resources verified in this project:

Component	Technology
Programming Language	Python
Machine Learning Library	scikit-learn
Deep Learning Framework	PyTorch / pytorch-tabnet
Optical Character Recognition	EasyOCR
Data Manipulation	Pandas / NumPy
Explainability Engine	SHAP (SHapley Additive exPlanations)
Dashboard / User Interface	Streamlit
Ontology Databases	Human Phenotype Ontology (hp.obo / phenotype.hpoa)
Document Generation	ReportLab

Reproducibility and Audit Trails: All model artifacts (such as ``rf_model.joblib``) are stored in ``models/``. The cohort generation is driven by ``preprocessing/generate_cohort.py`` with reproducible random seeds. Training and validation scripts are maintained inside ``training/``. Researchers can recreate the environment using ``requirements.txt`` and run ``verify_scientific.py`` to audit and reproduce all benchmarks.

20. Final Project Status & Conclusion

The following table reviews the current implementation status of all project components:

Component	Status
HPO Preprocessing	Completed
Synthetic Cohort Generation	Completed
Feature Engineering	Completed
Baseline ML Models	Completed
Random Forest Classifier	Completed
TabNet Benchmark	Completed
SHAP Explainability	Completed
Streamlit Application	Completed
OCR Pipeline	Implemented
OCR Accuracy	Requires further improvement
Real-world Clinical Validation	Future work

Scientific Conclusion

The project successfully demonstrates an HPO-based rare disease prediction prototype using machine learning, deep-learning benchmarking, explainability, and image-based clinical record processing. The current calibrated Random Forest classifier achieves 97.78% test accuracy on the synthetic HPO-derived evaluation cohort, offering high diagnostic predictability in simulated settings. However, these results are bounded by the synthetic evaluation framework and require independent validation using real-world clinical data before any actual deployment in medical environments.